Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If α, β, γ are the roots of $x^3 + ax^2 + b = 0$, then the value of $\begin{vmatrix} \alpha & \beta & \gamma \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{vmatrix}$ is

(A) $-a^3$

(B) $a^3 - 3b$

(C) a^3

(D) $a^2 - 3b$

Q.2

The parameter, on which the value of the determinant does not depend upon is

$$\begin{vmatrix} 1 & a & a^2 \\ \cos(p-d)x & \cos px & \cos(p+d)x \\ \sin(p-d)x & \sin px & \sin(p+d)x \end{vmatrix}$$

(A) a

(B) p

(C) d

(D) x

Q.3

If $D = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1+x & 1 \\ 1 & 1 & 1+y \end{vmatrix}$ for $x \neq 0, y \neq 0$ then D is

(A) divisible by neither x nor y

(B) divisible by both x and y

(C) divisible by x but not y

(D) divisible by y but not x

Q.4

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If the system of equations $x+y+z=0, \quad ax+by+z=0$ and $bx+y+z=0$ has a non-trivial solution, then

(A) $b^2=2b-1$

(B) $b-a=0$

(C) $b^2=2b$

(D) $b^2=2b+1$

Q.5

If $\begin{vmatrix} x^n & x^{n+2} & x^{n+3} \\ y^n & y^{n+2} & y^{n+3} \\ z^n & z^{n+2} & z^{n+3} \end{vmatrix} = (x-y)(y-z)(z-x)\left(\frac{1}{x}+\frac{1}{y}+\frac{1}{z}\right)$, then n equals

(A) 1

(B) -1

(C) 2

(D) -2

Q.6

Let $f(x)=x(x-1)$, then $\Delta = \begin{vmatrix} f(0) & f(1) & f(2) \\ f(1) & f(2) & f(3) \\ f(2) & f(3) & f(4) \end{vmatrix}$ is equal to

(A) $-2!$

(B) $-3!-2!$

(C) 0

(D) none of these

Q.7

The value of the determinant $\begin{vmatrix} ka & k^2+a^2 & 1 \\ kb & k^2+b^2 & 1 \\ kc & k^2+c^2 & 1 \end{vmatrix}$ is

(A) $k(a+b)(b+c)(c+a)$

(B) $kabc\left(a^2+b^2+c^2\right)$

(C) $k(a-b)(b-c)(c-a)$

(D) $k(a+b-c)(b+c-a)(c+a-b)$

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Q.8

If the system of equations $ax + y + z = 0, x + by + z = 0$ and $x + y + cz = 0$, where $a, b, c \neq 1$ has a non trivial solution, then the value of $\frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c}$ is equal to

(A) -1

(B) 0

(C) 1

(D) none of these

Q.9

If $x = -9$ is a root of the equation $\begin{vmatrix} x & 3 & 7 \\ 2 & x & 2 \\ 7 & 6 & x \end{vmatrix} = 0$ then the other two roots are

(A) $(3, 7)$

(B) $(2, 7)$

(C) $(3, 6)$

(D) $(2, 6)$

Q.10

The value of λ for which the system of equations $2x - y - z = 12, x - 2y + z = -4$ and $x + y + \lambda z = 4$ has no solution is

(A) 3

(B) -3

(C) 2

(D) -2

Q.11

If $a + b + c > 0$, and $\Delta = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ then

(A) $\Delta < 0$

(B) $\Delta \leq 0$

(C) $\Delta > 0$

(D) $\Delta = 0$

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Q.12

The number of values of k for which the system of equations

$$(k+1)x+8y=4k$$
$$kx+(k+3)y=3k-1$$

has infinitely many solution is

(A) 0

(B) 1

(C) 2

(D) infinite

Q.13

If 'ω' is a non real cube root of unit. Then $\begin{vmatrix} 1 & 1+i+\omega^2 & \omega^2 \\ 1-i & -1 & \omega^2-1 \\ -i & -i+\omega-1 & -1 \end{vmatrix}$, equal to

(A) 1

(B) i

(C) ω

(D) zero

Q.14

If $2ax-2y+3z=0, x+ay+2z=0$ and $2x+az=0$ have a non-trivial solution, then

(A) $a=2$

(B) $a=1$

(C) $a=0$

(D) none of these

Q.15

If $\Delta_a=\begin{vmatrix} a-1 & n & 6 \\ (a-1)^2 & 2n^2 & 4n-2 \\ (a-1)^3 & 3n^3 & 3n^2-3n \end{vmatrix}$, then $\sum_{a=1}^{n}\Delta_a$ is equal to

(A) 0

(B) 1

(C) $\left(\frac{n(n+1)}{2}\right)\left(\frac{a(a+1)}{2}\right)$

(D) none of these

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Q.16

If in a $\Delta ABC, \sin^3 A + \sin^3 B + \sin^3 C = 3 \sin A \cdot \sin B \cdot \sin C$, then the value of the determinant $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ is

(A) 0

(B) $(a+b+c)^3$

(C) $(a+b+c)(ab+bc+ca)$

(D) none of these

Q.17

If $a \in \mathbf{R}$ and the system of equations $x + ay = 0, az + y = 0, ax + z = 0$ has infinite number of solution then the value of a is

(A) -1

(B) 1

(C) 0

(D) no real value

Q.18

Let $f(\theta) = \begin{vmatrix} 1 & \cos\theta & -1 \\ -\sin\theta & 1 & -\cos\theta \\ -1 & \sin\theta & 1 \end{vmatrix}$

Suppose A and B are respectively maximum and minimum value of $f(\theta)$. Then (A, B) is equal to:

(A) $(2, 1)$

(B) $(2, 0)$

(C) $(\sqrt{2}, 1)$

(D) $\left(2, \frac{1}{\sqrt{2}}\right)$

Q.19

If the system of linear equations

$x + 2ay + az = 0$

$x + 3by + bz = 0$

$x + 4cy + cz = 0$

has a non-zero solution, then a, b, c

(A) are in G.P.

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(B) are in H.P.

(C) satisfy $a+2b+3c=0$

(D) are in A.P.

Q.20

If $f(x), g(x)$ and $h(x)$ are three polynomials of degree 3 then $\phi(x)=\begin{vmatrix} f'(x) & g'(x) & h'(x) \\ f''(x) & g''(x) & h''(x) \\ f'''(x) & g'''(x) & h'''(x) \end{vmatrix}$ is a polynomial of degree

(A) 3

(B) 4

(C) 5

(D) none of these

Q.21

The modulus of the value of $\begin{vmatrix} 1+i & 1-i & i \\ 1-i & i & 1+i \\ i & 1+i & 1-i \end{vmatrix}$ is equal to $\sqrt{k}$ then $k=$

Q.22

$D_r=\begin{vmatrix} 2^{r-1} & 2.3^{r-1} & 4.5^{r-1} \\ x & y & z \\ 2^n-1 & 3^n-1 & 5^n-1 \end{vmatrix}$, then the value of $\sum_{r=1}^{n} D_r$ is equal to

Q.23

If $p\lambda^4+q\lambda^3+r\lambda^2+s\lambda+t=\begin{vmatrix} \lambda^2+3\lambda & \lambda-1 & \lambda+3 \\ \lambda^2+1 & 2-\lambda & \lambda-3 \\ \lambda^2-3 & \lambda+4 & 3\lambda \end{vmatrix}$, then $|p|$ is equal to

Q.24

If $f(x)=\begin{vmatrix} \sin^2\theta & \cos^2\theta & x \\ \cos^2\theta & x & \sin^2\theta \\ x & \sin^2\theta & \cos^2\theta \end{vmatrix}$ $\theta \in \left(0, \frac{\pi}{2}\right)$, then roots of the number of roots of $f(x)=0$ is

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Q.25

If α, β, γ are three real numbers such that $\alpha + \beta + \gamma = 0$, then

$$\Delta = \begin{vmatrix} 1 & \cos\gamma & \cos\beta \\ \cos\gamma & 1 & \cos\alpha \\ \cos\beta & \cos\alpha & 1 \end{vmatrix} \text{ equals}$$

Q.26

Let $\Delta(x) = \begin{vmatrix} 3+2\sin^4 x & 2\cos^4 x & \sin^2 2x \\ 2\sin^4 x & 3+2\cos^4 x & \sin^2 2x \\ 2\sin^4 x & 2\cos^4 x & 3+\sin^2 2x \end{vmatrix}$ then $\int_{-\pi/2}^{\pi/2} x\Delta(x)dx$ equals

Q.27

The integral values of λ for which the system of equations

$$\begin{aligned} x + y - 3 &= 0 \\ (1+\lambda)x + (2+\lambda)y - 8 &= 0 \\ x - (1+\lambda)y + (2+\lambda) &= 0 \end{aligned}$$

has a non-trivial solution, are

Q.28

Let $p\lambda^4 + q\lambda^3 + r\lambda^2 + s\lambda + t$

$$= \begin{vmatrix} \lambda^2+3\lambda & \lambda-1 & \lambda-3 \\ \lambda-1 & -2\lambda & \lambda-4 \\ \lambda-3 & \lambda+4 & 3\lambda \end{vmatrix}$$

where p, q, r, s and t are constants. Then value of t is

Q.29

Suppose A, B, C are angles of a triangle, and let

$$\Delta = \begin{vmatrix} e^{2iA} & e^{-iC} & e^{-iB} \\ e^{-iC} & e^{2iB} & e^{-iA} \\ e^{-iB} & e^{-iA} & e^{2iC} \end{vmatrix}$$

Then value of Δ^2 is

Q.30

Suppose $a, b, c, \alpha \in \mathbf{R}$ and $abc\alpha \neq 0$. If the system of equations:

$(a+\alpha)x + \alpha y + \alpha z = 0$ ...(i)

$\alpha x + (b+\alpha)y + \alpha z = 0$ ...(ii)

$\alpha x + \alpha y + (\alpha + c)z = 0$ ...(iii)

has a non-trivial solution, then the modulus A(1/a+1/b+1/c) is

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Q.1 Answer:

a^3

Solution:

Since α, β, γ are the roots of the given equations, therefore $\alpha+\beta+\gamma=-\mathrm{a}, \alpha\beta+\beta\gamma+\gamma\alpha=0$ and $\alpha\beta\gamma=-b$.

$$\text{Now, } \begin{vmatrix} \alpha & \beta & \gamma \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{vmatrix} = -(\alpha+\beta+\gamma)\left(\alpha^2+\beta^2+\gamma^2-\alpha\beta-\beta\gamma-\gamma\alpha\right)$$

$$= -(\alpha+\beta+\gamma)\left\{(\alpha+\beta+\gamma)^2-3(\alpha\beta+\beta\gamma+\gamma\alpha)\right\}$$

$$= -(-a)\left\{(-a)\wedge 2-3\times 0\right\}$$

$$= a^3$$

Q.2 Answer:

p

Solution:

$$\text{Let } \Delta = \begin{vmatrix} 1 & a & a^2 \\ \cos(p-d)x & \cos px & \cos(p+d)x \\ \sin(p-d)x & \sin px & \sin(p+d)x \end{vmatrix}$$

Applying $C_1 \to C_1 + C_3$

$$\Rightarrow \Delta = \begin{vmatrix} 1+a^2 & a & a^2 \\ \cos(p-d)x+\cos(p+d)x & \cos px & \cos(p+d)x \\ \sin(p-d)x+\sin(p+d)x & \sin px & \sin(p+d)x \end{vmatrix}$$

$$\Rightarrow \Delta = \begin{vmatrix} 1+a^2 & a & a^2 \\ 2\cos px\cos dx & \cos px & \cos(p+d)x \\ 2\sin px\cos dx & \sin px & \sin(p+d)x \end{vmatrix}$$

Applying $C_1 \to C_1 - 2\cos dx C_2$

$$\Rightarrow \Delta = \begin{vmatrix} 1+a^2-2a\cos dx & a & a^2 \\ 0 & \cos px & \cos(p+d)x \\ 0 & \sin px & \sin(p+d)x \end{vmatrix}$$

$$\Rightarrow \Delta = \left(1+a^2-2a\cos dx\right)\left[\sin(p+d)x\cos px-\sin px\cos(p+d)x\right]$$

$$\Rightarrow \Delta = \left(1+a^2-2a\cos dx\right)\sin dx$$

Which is independent of p.

Q.3 Answer:

divisible by both x and y

Solution:

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$$D=\begin{vmatrix}1 & 1 & 1\\ 1 & 1+x & 1\\ 1 & 1 & 1+y\end{vmatrix} \qquad \begin{matrix}C_3 \to C_3 - C_1\\ C_2 \to C_2 - C_1\end{matrix}$$

$$=\begin{vmatrix}1 & 0 & 0\\ 1 & x & 0\\ 1 & 0 & y\end{vmatrix} = xy$$

Q.4 Answer:

$b^2 = 2b - 1$

Solution:

$$\begin{vmatrix}1 & 1 & 1\\ a & b & 1\\ b & 1 & 1\end{vmatrix} = 0 \Rightarrow b^2 - 2b + 1 = 0 \Rightarrow b^2 = 2b - 1$$

Q.5 Answer:

-1

Solution:

The degree of the determinant is $n + (n+2) + (n+3) = 3n + 5$ and the degree of the expression on RHS is 2.

$\therefore \quad 3n + 5 = 2 \quad \Rightarrow \quad n = -1$

Q.6 Answer:

$-3! - 2!$

Solution:

$f(x) = x(x-1)$

$$\therefore \quad \Delta = \begin{vmatrix}0 & 0 & 2\\ 0 & 2 & 6\\ 2 & 6 & 12\end{vmatrix} = 2\begin{vmatrix}0 & 2\\ 2 & 6\end{vmatrix} = 2(0-4) = -8 = -6 - 2 = -3! - 2!$$

Q.7 Answer:

$k(a-b)(b-c)(c-a)$

Solution:

$$\begin{vmatrix}ka & k^2+a^2 & 1\\ kb & k^2+b^2 & 1\\ kc & k^2+c^2 & 1\end{vmatrix} = k\begin{vmatrix}a & k^2+a^2 & 1\\ b & k^2+b^2 & 1\\ c & k^2+c^2 & 1\end{vmatrix}$$

$$= k\left\{\begin{vmatrix}a & k^2 & 1\\ b & k^2 & 1\\ c & k^2 & 1\end{vmatrix} + \begin{vmatrix}a & a^2 & 1\\ b & b^2 & 1\\ c & c^2 & 1\end{vmatrix}\right\} = k.k^2\begin{vmatrix}a & 1 & 1\\ b & 1 & 1\\ c & 1 & 1\end{vmatrix} + k\begin{vmatrix}1 & a & a^2\\ 1 & b & b^2\\ 1 & c & c^2\end{vmatrix}$$

$$= k^3.0 + k(a-b)(b-c)(c-a)$$

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Q.8 Answer:

1

Solution:

$$\begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & c \end{vmatrix} = 0 \qquad \begin{matrix} R_2 \to R_2 - R_1 \\ R_3 \to R_3 - R_1 \end{matrix}$$

$$\begin{vmatrix} a & 1 & 1 \\ 1-a & b-1 & 0 \\ 1-a & 0 & c-1 \end{vmatrix} = 0$$

Expanding, $a(b-1)(c-1) - (1-a)(b-1) - (1-a)(c-1) = 0$

Dividing by $(1-a)(1-b)(1-c)$**, we get** $\frac{a}{1-a} + \frac{1}{1-c} + \frac{1}{1-b} = 0$

$\frac{a}{1-a} + \frac{1}{1-a} + \frac{1}{1-c} + \frac{1}{1-b} = \frac{1}{1-a}$

$\Rightarrow \quad \frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c} = \frac{1}{1-a} - \frac{a}{1-a} = \frac{1-a}{1-a} = 1$

Q.9 Answer:

$(2, 7)$

Solution:

Given $\begin{vmatrix} x & 3 & 7 \\ 2 & x & 2 \\ 7 & 6 & x \end{vmatrix} = 0 \quad \Rightarrow x\left(x^2 - 12\right) - 3(2x - 14) + 7(12 - 7x) = 0$

$\Rightarrow \quad x^3 - 67x + 126 = 0$

$x = -9$ **is one root of this equation, let other two roots are** α **and** β

$\Rightarrow \quad (x+9)(x-\alpha)(x-\beta) = x^3 - 67x + 126$

$(x-\alpha)(x-\beta) = \frac{x^3 - 67x + 126}{x+9} = x^2 - 9x + 14$

$\Rightarrow \quad (x-\alpha)(x-\beta) = (x-2)(x-7) = 0 \Rightarrow$ **Other two roots are** $(2, 7)$

Q.10 Answer:

-2

Solution:

The coefficient determinant $D = \begin{vmatrix} 2 & -1 & -1 \\ 1 & -2 & 1 \\ 1 & 1 & \lambda \end{vmatrix} = 3\lambda - 6$

For no solution, the necessary condition is $D = 0$

i.e., $-3\lambda - 6 = 0 \Rightarrow \lambda = -2$

Q.11 Answer:

$\Delta \leq 0$

Solution:

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Applying $C_1 \to C_1 + C_2 + C_3$ we get

$$\Delta = \begin{vmatrix} a+b+c & b & c \\ a+b+c & c & a \\ a+b+c & a & b \end{vmatrix} = (a+b+c)\begin{vmatrix} 1 & b & c \\ 1 & c & a \\ 1 & a & b \end{vmatrix}$$

Applying $R_2 \to R_2 - R_1$ and $R_3 \to R_3 - R_1$ we get

$$(a+b+c)\begin{vmatrix} 1 & b & c \\ 0 & c-b & a-c \\ 0 & a-b & b-c \end{vmatrix}$$

Expanding along with C_1 we get

$$\Delta = (a+b+c)\left[-(b-c)^2 - (a-b)(a-c)\right]$$
$$= -(a+b+c)\left(b^2+c^2-2bc+a^2-ab-ac+bc\right)$$
$$= -\tfrac{1}{2}(a+b+c)\left[\left(b^2+c^2-2bc\right)+\left(c^2+a^2-2ca\right)+\left(a^2+b^2-2ab\right)\right]$$
$$= -\tfrac{1}{2}(a+b+c)\left[(b-c)^2+(c-a)^2+(a-b)^2\right] \le 0$$
$$= [\because a+b+c > 0]$$

Q.12 Answer:

1

Solution:

Let $\Delta = \begin{vmatrix} k+1 & 8 \\ k & k+3 \end{vmatrix}$

$= \mathbf{k^2 + 4k + 3 = (k-1)(k-3)}$

For the system of equation to have no solution $\Delta = 0 \Rightarrow \mathrm{k} = 3, 1$

For $\mathrm{k} = 1$**, the system of equation becomes** $2x + 8y = 4$ **and** $x + 4y = 2$

$\Rightarrow \mathrm{x + 4y = 2}, \quad \mathrm{x + 4y = 2}$

This system has infinite number of solution

For $\mathbf{k} = 3$**. the system of equation becomes** $4x + 8y = 12$ **and** $3x + 6y = 8$

$\Rightarrow x + 2y = 3, x + 2y = \frac{8}{3}$

Thus, in this case the system of equation has no solution.

The number of values of k **for which the given system of equations has no solution is 1.**

Q.13 Answer:

zero

Solution:

Applying $R_1 \to R_1 - (R_2 - R_3)$ **we get,**

$$0 = \begin{vmatrix} 0 & 1+\omega^2+\omega & 0 \\ 1-i & -1 & \omega^2-1 \\ -i & -i+\omega-1 & -1 \end{vmatrix}$$

$$= \begin{vmatrix} 0 & 0 & 0 \\ 1-i & -1 & \omega^2-1 \\ -i & -i+\omega-1 & -1 \end{vmatrix} = 0$$

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Q.14 Answer:

$a = 2$

Solution:

For no trivial solution, we must have, $\begin{vmatrix} 2a & -2 & 3 \\ 1 & a & 2 \\ 2 & 0 & a \end{vmatrix} = 0$

$\Rightarrow \quad 2a\left(a^2 - 0\right) + 2(a-4) + 3(0-2a) = 0$

$\Rightarrow \quad 2a^3 + 2a - 8 + 0 - 6a = 0 \Rightarrow 2a^3 - 4a - 8 = 0 \Rightarrow a^3 - 2a - 4 = 0$

$\Rightarrow \quad a^3 - 2a^2 + 2a^2 - 4a + 2a - 4 = 0$

$\Rightarrow \quad a^2(a-2) + 2a(a-2) + 2(a-2) = 0 \Rightarrow (a-2)\left(a^2 + 2a + 2\right) = 0 \Rightarrow a = 2$

Q.15 Answer:

0

Solution:

$$\sum_{a=1}^{n} \Delta_a = \begin{vmatrix} \frac{n(n-1)}{2} & n & 6 \\ \frac{n(n-1)(2(n-1)+1)}{6} & 2n^2 & 4n-2 \\ \left[\frac{n(n-1)}{2}\right]^2 & 3n^3 & 3n^2 - 3n \end{vmatrix}$$

$$= \frac{n(n-1)}{2\times 6} \begin{vmatrix} 6 & n & 6 \\ 4n-2 & 2n^2 & 4n-2 \\ 3n^2 - n & 3n^3 & 3n^2 - 3n \end{vmatrix} = 0$$

Q.16 Answer:

0

Solution:

$a^3 + b^3 + c^3 = 3abc$ **[By sine rule]**

$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = a^3 + b^3 + c^3 - 3abc = 0$$

Q.17 Answer:

-1

Solution:

We have $x = -ay$,

therefore $-a^2y + z = 0 \Rightarrow z = a^2y$.

Thus, $az + y = 0 \Rightarrow \left(a^3 + 1\right)z = 0$.

For the system of equations to have infinite number of solutions $a^3 + 1 = 0$

$\Rightarrow \quad a = -1 \quad [\because a$ **is real]**

Q.18 Answer:

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$(2, 0)$

Solution:

Using $C_1 \to C_1 + C_3$, we get

$$f(\theta) = \begin{vmatrix} 0 & \cos\theta & -1 \\ -(\sin\theta + \cos\theta) & 1 & -\cos\theta \\ 0 & \sin\theta & 1 \end{vmatrix}$$

Evaluating along C_1, we get

$$\begin{aligned} f(\theta) &= (\sin\theta + \cos\theta) \begin{vmatrix} \cos\theta & -1 \\ \sin\theta & 1 \end{vmatrix} \\ &= (\sin\theta + \cos\theta)^2 \\ &= \left[\sqrt{2}\left(\frac{1}{\sqrt{2}}\sin\theta + \frac{1}{\sqrt{2}}\cos\theta\right)\right]^2 \\ &= 2\sin^2\left(\frac{\pi}{4} + \theta\right) \end{aligned}$$

As $0 \le \sin^2\left(\frac{\pi}{4} + \theta\right) \le 1$, we get

$0 \le f(\theta) \le 2$

$A = 2$ **for** $\theta = \frac{\pi}{4}$

and $B = 0$ **for** $\theta = -\frac{\pi}{4}$

Thus, $(A, B) = (2, 0)$

Q.19 Answer:

are in H.P.

Solution:

The system will have a non-zero solution if

$$\Delta = \begin{vmatrix} 1 & 2a & a \\ 1 & 3b & b \\ 1 & 4c & c \end{vmatrix} = 0$$

Using $C_2 \to C_2 - 2C_3$, we get

$$\Delta = \begin{vmatrix} 1 & 0 & a \\ 1 & b & b \\ 1 & 2c & c \end{vmatrix} = 0$$

$\Rightarrow 1(bc - 2bc) + a(2c - b) = 0 \Rightarrow 2ac = bc + ab$

$\Rightarrow \frac{2ac}{a+c} = b \Rightarrow a, b, c$ are in H.P.

Q.20 Answer:

none of these

Solution:

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We have

$$\phi'(x) = \begin{vmatrix} f''(x) & g''(x) & h''(x) \\ f''(x) & g''(x) & h''(x) \\ f'''(x) & g'''(x) & h'''(x) \end{vmatrix} + \begin{vmatrix} f'(x) & g'(x) & h'(x) \\ f'''(x) & g'''(x) & h'''(x) \\ f'''(x) & g'''(x) & h'''(x) \end{vmatrix} + \begin{vmatrix} f'(x) & g'(x) & h'(x) \\ f''(x) & g''(x) & h''(x) \\ f^{iv}(x) & g^{iv}(x) & h^{iv}(x) \end{vmatrix}$$

$$= 0 + 0 + \begin{vmatrix} f'(x) & g'(x) & h'(x) \\ f''(x) & g''(x) & h''(x) \\ 0 & 0 & 0 \end{vmatrix} = 0$$

since f, g, h are polynomials of degree 3, $f^{iv}(x) = g^{iv}(x) = h^{iv}(x) = 0$

$\Rightarrow \phi(x)$ must be a constant.

Q.21 Answer:

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Solution:

$$\Delta = (2+i)\begin{vmatrix} 1 & 1 & i \\ 1 & 1+2i & 1+i \\ 1 & 2 & 1-i \end{vmatrix}$$

$$(2+i)\begin{vmatrix} 0 & -2i & -1 \\ 0 & -1+2i & 2i \\ 1 & 2 & 1-i \end{vmatrix} \quad \text{by} \quad \begin{matrix} R_1 \to R_1 - R_2 \\ R_2 \to R_2 - R_3 \end{matrix}$$

$$= (2+i)\left\{-4i^2 + (-1+2i)\right\} = (2+i)(4-1+2i) = (2+i)(3+2i) = 4+7i$$

Q.22 Answer:

—

Solution:

$$D_r = \begin{vmatrix} 2^{r-1} & 2.3^{r-1} & 4.5^{r-1} \\ x & y & z \\ 2^n - 1 & 3^n - 1 & 5^n - 1 \end{vmatrix}$$

$$\Rightarrow \sum_{r=1}^{n} D_r = \begin{vmatrix} \sum_{r=1}^{n} 2^{r-1} & \sum_{r=1}^{n} 2.3^{r-1} & \sum_{r=1}^{n} 4.5^{r-1} \\ x & y & z \\ 2^n - 1 & 3^n - 1 & 5^n - 1 \end{vmatrix}$$

$$\Rightarrow \sum_{r=1}^{n} D_r = \begin{vmatrix} 2^n - 1 & 3^n - 1 & 5^n - 1 \\ x & y & z \\ 2^n - 1 & 3^n - 1 & 5^n - 1 \end{vmatrix}$$

Since we know that $\sum_{r=1}^{n} 2^{r-1} = \frac{2^n - 1}{2-1} = 2^n - 1$

$2\sum_{r=1}^{n} 3^{r-1} = 2\frac{3^n - 1}{3-1} = 3^n - 1$ **and** $4\sum_{r=1}^{n} 5^{r-1} = 4\frac{5^n - 1}{5-1} = 5^n - 1$

$\Rightarrow \quad \sum_{r=1}^{n} D_r = 0 \quad (\because R_1 \equiv R_3)$

Q.23 Answer:

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Solution:

We divide LHS by λ^4 and C_1 by λ^2, C_2 by λ and C_3 by λ on the RHS to obtain,

$$p+q\left(\frac{1}{\lambda}\right)+r\left(\frac{1}{\lambda}\right)^2+s\left(\frac{1}{\lambda}\right)^3+t\left(\frac{1}{\lambda}\right)^4=\begin{vmatrix}1+3/\lambda & 1-1/\lambda & 1+3/\lambda\\ 1+1/\lambda^2 & 2/\lambda-1 & 1-3/\lambda\\ 1-3/\lambda^2 & 1+4/\lambda & 3\end{vmatrix}$$

Taking limit as $\lambda\to\infty$, we get

$$p=\begin{vmatrix}1 & 1 & 1\\ 1 & -1 & 1\\ 1 & 1 & 3\end{vmatrix}=\begin{vmatrix}1 & 1 & 1\\ 0 & -2 & 0\\ 0 & 0 & 2\end{vmatrix}=-4 \quad \text{[using } R_2\to R_2-R_1, R_3\to R_3-R_1\text{]}$$

Q.24 Answer:

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Solution:

$$f(x)=\left(\sin^2\theta+\cos^2\theta+x\right)\begin{vmatrix}1 & \cos^2\theta & x\\ 1 & x & \sin^2\theta\\ 1 & \sin^2\theta & \cos^2\theta\end{vmatrix}=(x+$$

$$1)\begin{vmatrix}1 & \cos^2\theta & x\\ 0 & x-\cos^2\theta & \sin^2\theta-x\\ 0 & \sin^2\theta-\cos^2\theta & \cos^2\theta-x\end{vmatrix}$$

$$=(x+1)\left[\left(x-\cos^2\theta\right)\left(\cos^2\theta-x\right)-\left(\sin^2\theta-\cos^2\theta\right)\left(\sin^2\theta-x\right)\right]$$

$$=(x+1)\left[-x^2-\cos^4\theta+2x\cos^2\theta-x\cos^2\theta+x\sin^2\theta-\sin^4\theta+\sin^2\theta\cos^2\theta\right]$$

$$=\left(-\tfrac{1}{2}\right)(x+1)\left[\left(x-\sin^2\theta\right)^2+\left(x-\cos^2\theta\right)^2+\left(\sin^2\theta-\cos^2\theta\right)^2\right]$$

So, $f(x)=0$ if $x=-1$ or $x=\sin^2\theta=\cos^2\theta$

$\sin^2\theta=\cos^2\theta\Rightarrow\quad\theta=\frac{\pi}{4}\Rightarrow x=\frac{1}{2}.$

Hence $x=-1,\frac{1}{2}$

Q.25 Answer:

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Solution:

Let A, B and C be three real numbers such that $\alpha=B-C, \beta=C-A$ and $\gamma=A-B$, clearly $\alpha+\beta+\gamma=0$. We have

$$\Delta=\begin{vmatrix}1 & \cos(A-B) & \cos(C-A)\\ \cos(A-B) & 1 & \cos(B-C)\\ \cos(C-A) & \cos(B-C) & 1\end{vmatrix}$$

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$$\begin{vmatrix} \cos^2 A+\sin^2 A & \cos A\cos B+\sin A\sin B & \cos C\cos A+\sin A\sin C \\ \cos A\cos B+\sin A\sin B & \cos^2 B+\sin^2 B & \cos B\cos C+\sin B\sin C \\ \cos C\cos A+\sin C\sin A & \cos B\cos C+\sin B\sin C & \cos^2 C+\sin^2 C \end{vmatrix}$$

$$= \begin{vmatrix} \cos A & \sin A & 0 \\ \cos B & \sin B & 0 \\ \cos C & \sin C & 0 \end{vmatrix} \begin{vmatrix} \cos A & \sin A & 0 \\ \cos B & \sin B & 0 \\ \cos C & \sin C & 0 \end{vmatrix} = (0)(0) = 0$$

Q.26 Answer:

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Solution:

Using $C_1 \to C_1 + C_2 + C_3$, we get

$$\Delta(x) = f(x) \begin{vmatrix} 1 & 2\cos^4 x & \sin^2 2x \\ 1 & 3+2\cos^4 x & \sin^2 2x \\ 1 & 2\cos^4 x & 3+\sin^2 2x \end{vmatrix}$$

$$\text{where } f(x) = 3 + 2\sin^4 x + 2\cos^4 x + \sin^2 2x$$

$$= 3 + 2\left(\sin^4 x + \cos^4 x + 2\sin^2 x \cos^2 x\right)$$

$$= 3 + 2\left(\cos^2 x + \sin^2 x\right)^2 = 5$$

Applying $C_2 \to C_2 - \left(2\cos^4 x\right) C_1$ and $C_3 \to C_3 - \left(\sin^2 2x\right) C_1$, we get

$$\Delta(x) = 5 \begin{vmatrix} 1 & 0 & 0 \\ 1 & 3 & 0 \\ 1 & 0 & 3 \end{vmatrix} = 45$$

Thus, $\int_{-\pi/2}^{\pi/2} x\Delta(x)dx = 45\int_{-\pi/2}^{\pi/2} xdx = 0$

Q.27 Answer:

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Solution:

The given system of equations has a non trivial solution if

$$\Delta = \begin{vmatrix} 1 & 1 & -3 \\ 1+\lambda & 2+\lambda & -8 \\ 1 & -(1+\lambda) & 2+\lambda \end{vmatrix} = 0$$

Applying $C_2 \to C_2 - C_1$ and $C_3 \to C_3 + 3C_1$, we get

$$\Delta = \begin{vmatrix} 1 & 0 & 0 \\ 1+\lambda & 1 & -5+3\lambda \\ 1 & -2-\lambda & 5+\lambda \end{vmatrix} = 0$$

$$\Rightarrow (5+\lambda) + (2+\lambda)(3\lambda - 5) = 0$$

$$\Rightarrow 5 + \lambda + 6\lambda - 10 + 3\lambda^2 - 5\lambda = 0$$

$$\Rightarrow \quad 3\lambda^2 + 2\lambda - 5 = 0 \Rightarrow (3\lambda + 5)(\lambda - 1) = 0$$

$$\Rightarrow \lambda = -5/3 \text{ or } \lambda = 1.$$

Q.28 Answer:

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Solution:

Putting $\lambda = 0$, we obtain

$$t = \begin{vmatrix} 0 & -1 & 3 \\ 1 & 0 & -4 \\ -3 & 4 & 0 \end{vmatrix} = 0$$

as it is a skew symmetric determinant of odd order.

Q.29 Answer:

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Solution:

Taking e^{iA}, common from R_1, e^{iB} from R_2 and e^{iC} from R_3, we get

$\Delta = e^{i(A+B+C)}\Delta_1$

where

$$\Delta_1 = \begin{vmatrix} e^{iA} & e^{-i(A+C)} & e^{-i(A+B)} \\ e^{-i(B+C)} & e^{iB} & e^{-i(A+B)} \\ e^{-i(B+C)} & e^{-i(A+C)} & e^{iC} \end{vmatrix}$$

$$\text{But } A+B+C = \pi, \text{ so that } e^{i(A+B+C)} = e^{i\pi}$$

$$= \cos\pi + i\sin\pi = -1. \text{ Also ,}$$

$$A + C = \pi - B \quad \Rightarrow \quad e^{-i(A+C)} = e^{-\pi i}e^{iB} = -e^{iB}.$$

Thus, $\Delta_1 = \begin{vmatrix} e^{iA} & -e^{iB} & -e^{iC} \\ -e^{-iA} & e^{iB} & -e^{iC} \\ -e^{iA} & -e^{iB} & e^{iC} \end{vmatrix}$

$$= e^{i(A+B+C)} \begin{vmatrix} 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{vmatrix}$$

Using $C_1 \to C_1 + C_2$, we get

$$\Delta_1 = (-1)\begin{vmatrix} 0 & -1 & -1 \\ 0 & 1 & -1 \\ -2 & -1 & 1 \end{vmatrix} = (-1)(-2)(2) = 4$$

$$\text{Therefore, } \Delta = (-1)\Delta_1 = -4$$

Q.30 Answer:

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Solution:

From (1) and (2)

$ax - by = 0$

and from (2) and (3)

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$$ax - cz = 0$$

$$\therefore ax = by = cz$$

$$\Rightarrow \quad \frac{x}{1/a} = \frac{y}{1/b} = \frac{z}{1/c}$$

Putting in (1) we get

$\frac{a+\alpha}{a} + \frac{\alpha}{b} + \frac{\alpha}{c} = 0$

$\Rightarrow \alpha\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) = -1$

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